

GUSTAVUS, AK, USA

WMO: 703670

Lat: 58.425N

Lon: 135.707W

Elev: 12

StdP: 101.18

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25322

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.7	-14.0	-25.4	0.4	-14.3	-22.2	0.5	-10.2	11.0	4.4	9.3	3.9	0.7	350	0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.2	21.7	14.0	19.0	13.3	17.6	12.8	15.3	19.2	14.5	17.9	13.8	16.8	3.1	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.5	9.6	16.0	12.8	9.2	15.5	12.3	8.9	15.1	42.4	19.2	40.3	17.6	38.4	16.7	18.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.8	6.6	DB	-21.8	24.7	3.7	2.3	-24.4	26.4	-26.6	27.8	-28.6	29.1	-31.3	30.8	(4)
(5)			WB	-21.5	16.6	3.5	1.2	-24.0	17.4	-26.0	18.2	-28.0	18.9	-30.6	19.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-1.6	-1.2	0.6	4.4	8.3	11.7	13.4	13.3	10.1	5.7	1.0	-0.8	(6)	
	DBStd	6.57	5.40	4.75	3.97	2.54	2.47	2.13	1.66	1.76	2.18	2.77	4.35	5.26	(7)	
	HDD10.0	1966	359	313	292	168	63	5	0	1	24	135	271	336	(8)	
	HDD18.3	4708	617	546	550	418	310	200	154	158	248	392	521	594	(9)	
	CDD10.0	301	0	0	0	1	12	55	105	101	26	1	0	0	(10)	
	CDD18.3	1	0	0	0	0	0	1	1	0	0	0	0	0	(11)	
	CDH23.3	32	0	0	0	0	1	10	10	11	0	0	0	0	(12)	
	CDH26.7	3	0	0	0	0	0	1	2	1	0	0	0	0	(13)	
(14) Wind	WSAvg	2.3	2.0	2.3	2.7	2.5	2.3	1.9	1.7	1.8	2.1	2.6	2.5	2.6	(14)	
(15) Precipitation	PrecAvg	1814	188	148	131	92	75	61	85	117	216	259	221	220	(15)	
	PrecMax	2424	336	237	298	169	156	112	156	218	306	411	408	434	(16)	
	PrecMin	1474	83	51	21	25	18	23	25	44	77	119	72	115	(17)	
	PrecStd	226	62	54	56	36	35	20	33	44	48	67	72	87	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	8.0	11.2	17.5	21.8	25.0	24.1	24.8	19.9	13.0	10.0	9.1	(19)	
		MCWB	6.6	4.9	6.6	9.2	11.8	14.9	15.3	15.1	12.7	9.5	7.6	6.8	(20)	
	2%	DB	7.0	7.2	8.0	13.7	18.0	20.8	19.8	21.4	17.3	12.1	8.0	7.5	(21)	
		MCWB	5.0	4.6	4.7	7.2	10.5	13.4	14.6	14.6	11.4	9.1	6.0	5.6	(22)	
	5%	DB	5.9	6.1	7.1	11.4	16.1	17.8	17.9	18.4	15.2	11.1	7.3	6.2	(23)	
		MCWB	4.0	3.8	4.1	6.2	9.8	12.5	13.9	13.9	11.2	8.7	5.2	4.1	(24)	
	10%	DB	4.1	4.4	6.1	9.2	13.8	16.2	17.1	17.1	13.8	9.8	6.2	4.9	(25)	
		MCWB	2.9	2.6	3.4	5.5	8.7	11.7	13.4	13.4	11.0	7.9	4.5	3.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	5.7	7.4	10.0	12.7	15.8	16.3	16.6	13.9	10.7	8.0	7.4	(27)	
		MCDB	8.4	7.5	9.5	16.6	19.0	22.9	21.2	22.5	17.0	12.2	9.6	8.4	(28)	
	2%	WB	5.0	4.9	5.6	7.9	11.4	14.2	15.2	15.4	12.9	9.8	6.8	5.8	(29)	
		MCDB	6.8	6.8	7.7	12.1	16.8	19.2	18.7	19.3	15.5	11.1	7.8	7.5	(30)	
	5%	WB	4.1	4.1	4.7	6.8	10.4	13.0	14.4	14.5	12.1	8.9	5.6	4.6	(31)	
		MCDB	5.7	6.0	6.7	10.2	14.7	16.8	17.6	17.6	14.1	10.2	6.8	6.2	(32)	
	10%	WB	3.2	3.1	3.8	6.0	9.5	12.1	13.7	13.8	11.4	8.0	4.7	3.5	(33)	
		MCDB	4.2	4.4	5.6	8.9	13.4	15.5	16.5	16.5	13.3	9.1	6.2	4.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.9	7.4	8.9	9.0	7.5	6.2	6.7	6.8	6.0	5.5	5.2	(35)	
		MCDBR	5.2	6.5	8.7	14.1	14.6	12.2	9.7	11.9	11.5	7.6	5.2	5.4	(36)	
	5% WB	MCWBR	3.8	4.7	6.0	8.9	8.5	6.9	5.8	7.0	7.4	5.3	3.7	4.0	(37)	
		MCWBR	5.0	5.7	7.6	11.2	11.9	10.2	8.6	9.3	7.7	5.7	4.6	5.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.287	0.293	0.317	0.369	0.390	0.388	0.389	0.361	0.340	0.324	0.285	0.256	(40)	
		taud	2.144	2.326	2.297	2.203	2.254	2.349	2.392	2.499	2.537	2.435	2.217	2.110	(41)	
	Edn at Noon		525	745	828	833	840	848	839	842	807	701	529	437	(42)	
		Edh at Noon	50	68	92	120	122	113	107	89	74	61	46	36	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.39	1.13	2.36	3.90	5.00	5.06	4.36	3.51	2.24	1.18	0.52	0.26	(44)	
	RadStd		0.05	0.20	0.30	0.43	0.53	0.36	0.44	0.44	0.36	0.18	0.08	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page